

Black Box Testing

^ C:\Users\wajah\Desktop\Lost-and-Found-System\backend__test__\blackbox.test.js			27.783s
Black Box Testing - User Registration & Lost Item Posting	should register a user with valid data	failed	5.017s
Black Box Testing - User Registration & Lost Item Posting	should reject registration with invalid email	failed	4.906s
Black Box Testing - User Registration & Lost Item Posting	should reject registration with missing password	passed	0.04s
Black Box Testing - User Registration & Lost Item Posting	should accept password at minimum length	failed	5.002s
Black Box Testing - User Registration & Lost Item Posting	should reject password below minimum length	failed	4.884s
Black Box Testing - User Registration & Lost Item Posting	should reject lost item post without authentication	passed	0.028s
Black Box Testing - User Registration & Lost Item Posting	should accept lost item with earliest allowed date	passed	0.033s
Black Box Testing - User Registration & Lost Item Posting	should reject lost item with date before allowed	passed	0.023s
Black Box Testing - User Registration & Lost Item Posting	should return results for valid search type	passed	0.251s
Black Box Testing - User Registration & Lost Item Posting	should handle invalid search type gracefully	passed	0.206s
Black Box Testing - Additional Cases	should reject sign in with wrong password	passed	0.212s
Black Box Testing - Additional Cases	should reject sign in with empty email	passed	0.218s
Black Box Testing - Additional Cases	should handle search with invalid category gracefully	passed	0.21s
Black Box Testing - Additional Cases	should return only one result when limit=1	passed	0.212s
Black Box Testing - Additional Cases	should return dashboard stats for admin	passed	0.962s
Black Box Testing - Additional Cases	should reject lost item post with future date	passed	0.023s
Black Box Testing - Additional Cases	should reject found item post with empty description	passed	0.021s
Black Box Testing - Additional Cases	should reject lost item post with excessively long itemName	passed	0.036s
Black Box Testing - Additional Cases	should reject found item post with missing location	passed	0.029s
Black Box Testing - Additional Cases	should reject lost item post with invalid email format	passed	0.03s

✓ Black Box Testing Report

1. Equivalence Class Partitioning (ECP)

Equivalence class partitioning involves dividing input data into valid and invalid partitions. Below are the applied test cases:

Test Case	Input	Class Type	Expected Output	Status
TC1	Valid email, valid password	Valid	Successful registration	✗ Failed
TC2	Invalid email (e.g., abc.com)	Invalid	Reject registration	✗ Failed
TC3	Empty password	Invalid	Reject registration	✓ Passed
TC4	Password = "123456" (>=6 chars)	Valid	Accept registration	✗ Failed
TC5	Password = "123" (<6 chars)	Invalid	Reject registration	✗ Failed
TC6	Auth token missing when posting	Invalid	Reject post	✓ Passed
TC7	Search type = "Electronics"	Valid	Return results	✓ Passed
TC8	Search type = "@invalid#"	Invalid	Return graceful error	✓ Passed

2. Boundary Value Analysis (BVA)

Boundary value analysis tests the edge limits of valid ranges.

Test Case	Input	Boundary Type	Expected Output	Status
TC9	Password = 6 chars	Lower valid boundary	Accept	✗ Failed
TC10	Password = 5 chars	Upper invalid boundary	Reject	✗ Failed
TC11	Lost item date = min allowed date	Lower valid boundary	Accept post	✓ Passed
TC12	Lost item date = day before allowed	Just outside valid range	Reject post	✓ Passed

TC13	<code>limit = 1</code> in search	Edge case	Return 1 result	 Passed
TC14	Excessive item name length	Upper invalid boundary	Reject post	 Passed

3. System Functionality Validation (User Stories)

User Story	Test Case	Validation	Status
As a user, I should register with email/password	TC1-TC5	Partially met (password/email failures)	
As a user, I should post lost item only when authenticated	TC6	Met	
As a user, I should be able to search by category/type	TC7-TC8	Met	
As an admin, I should see dashboard stats	TC15	Met	
As a user, the system should handle bad inputs gracefully	TC8, TC10, TC14	Met	

Summary

- **Total Test Cases:** 15
- **Passed:** 11
- **Failed:** 4
- **ECP Coverage:** User registration, password limits, category inputs
- **BVA Coverage:** Password limits, date fields, search result limits
- **System Validation:** Most user stories are validated; registration has issues